

Objectives

For the second week, our group will focus on gathering the required set of data as well as the visualization and processing of said procured data. The total dataset that will be gathered must be from a verified source and at least consist of 500,000 rows to ensure lesser bias. We also planned to utilize the advantages offered by a Jupyter Notebook to streamline the phases of this week's process, allowing our group to work seamlessly on the project. This week was dedicated for large-scale data organization purposes while also displaying effective team collaboration.

Tasks Completed

For this week, my group assigned to me the task for Phase 1—Data Gathering and Procurement. This phase was simply for providing appropriate and reliable data in order to support the latter phases' processes. Below are the steps that I took in order to deliver my task:

1. Defining Requirements of Data — Our topic revolved in Urban Air Quality which is assessed by computing the AQI (Air Quality Index) of a certain site. Thus, this necessitated the dataset to include urban pollutant factors such as Sulfur Dioxide (SO₂) levels, Nitrogen Dioxide (NO₂) levels, Carbon Monoxide (CO) levels, and levels of Particulate Matter less than 2.5 micrometers (PM_{2.5}). The former set of pollutants also vary by time and location and as such the relevant dataset must include the location and the time at which the factors were read.
2. Identify reliable sources of datasets—The following were the main sites from which the datasets were retrieved: Kaggle, Google Dataset Search, [AQICN.org](https://aqicn.org).
3. Verify Licensing and Usage Rights.
4. Download — the csv files were downloaded and uploaded to a group shared folder.

Name	Owner	Date modified	File size
 Air Quality Data	 pcpm2023-9301-68233	23 Apr	—
 h.c.andersens-boulevard, copenhagen, denmark-air-quality.csv	 me	23 Apr me	123 KB
 seijo,-setagaya, tokyo, japan-air-quality.csv	 me	23 Apr me	129 KB
 beijing-air-quality.csv	 me	23 Apr me	142 KB
 charleroi,-pennsylvania-air-quality.csv	 me	23 Apr me	113 KB
 ottawa-air-quality.csv	 me	23 Apr me	112 KB
 bangkok-air-quality.csv	 me	23 Apr me	130 KB
 wilhelmsburg,-hamburg-air-quality.csv	 me	23 Apr me	117 KB
 helsinki-air-quality.csv	 me	23 Apr me	115 KB
 bakersfield-california-avenue, kern, california-air-q...	 me	23 Apr me	123 KB
 los-angeles-north main street-air-quality.csv	 me	23 Apr me	134 KB
 seoul-air-quality.csv	 me	23 Apr me	130 KB

5. Data Screening and Validation — the csv files were further screened in order to determine their accuracy and completeness.
6. Final Selection—the dataset with the largest repository of rows and least empty spaces were selected for processing in the next phases.

Tasks to be Completed

Data procurement has been fully accomplished with no other task for this week.

Results and Analysis

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date,sitename,county,aqi,pollutant,status,so2,co,o3,o3_8hr,pm10,pm2.5,no2,nox,no,windspeed,winddirec,unit,co_8hr,pm2.5_avg,pm10_avg,so2_avg,1
2024-08-31 23:00,Hukou,Hsinchu County,62.0,PM2.5,Moderate,0.9,0.17,35.0,40.2,18.0,17.0,2.3,2.6,0.3,2.3,225,,0.2,20.1,26.0,1.0,121.03886894,24
2024-08-31 23:00,Zhongming,Taichung City,50.0,,Good,1.6,0.32,27.9,35.1,27.0,14.0,7.6,9.3,1.6,1.1,184,,0.2,15.3,23.0,1.0,120.641092,24.151958,
2024-08-31 23:00,Zhudong,Hsinchu County,45.0,,Good,0.4,0.17,25.1,40.6,21.0,13.0,2.9,4.1,1.1,0.4,210,,0.2,13.8,24.0,0.0,121.08895493,24.740914
2024-08-31 23:00,Hsinchu,Hsinchu City,42.0,,Good,0.8,0.2,30.0,35.9,19.0,10.0,4.0,4.8,0.7,1.9,239,,0.2,13.0,26.0,1.0,120.97236752,24.8056356,2
2024-08-31 23:00,Toufen,Miaoli County,50.0,,Good,1.0,0.16,33.5,35.9,18.0,14.0,1.8,3.1,1.2,1.8,259,,0.1,15.3,28.0,1.0,120.89869286,24.69690679
2024-08-31 23:00,Miaoli,Miaoli County,40.0,,Good,1.1,0.17,35.2,35.0,15.0,12.0,4.0,5.1,1.1,1.4,235,,0.1,12.2,17.0,1.0,120.82011468,24.56499183
2024-08-31 23:00,Sanyi,Miaoli County,39.0,,Good,0.9,0.18,35.3,42.9,14.0,9.0,2.4,3.1,0.7,0.6,203,,0.1,11.4,16.0,0.0,120.75956754,24.38248443,2
2024-08-31 23:00,Fengyuan,Taichung City,44.0,,Good,1.3,0.24,24.6,39.7,21.0,12.0,6.8,7.3,0.5,1.2,38,,0.2,13.5,21.0,1.0,120.74252414,24.2569973
2024-08-31 23:00,Shalu,Taichung City,46.0,,Good,2.5,0.2,30.3,40.4,33.0,16.0,7.3,7.7,0.3,0.5,8,,0.1,14.2,26.0,1.0,120.568794,24.225628,29.0
2024-08-31 23:00,Dali,Taichung City,49.0,,Good,0.7,0.24,29.4,37.0,20.0,12.0,5.6,6.3,0.7,0.9,97,,0.2,15.1,21.0,0.0,120.67844444,24.09961111,30
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The final selection consisted of over five million rows. It displayed minimal empty spaces compared to other retrieved datasets with a validated source as cited by the dataset's author. The dataset meets all the required data needed for the development of the project.

Challenges and Solutions

- The main challenge for this task was verifying the trustworthiness of the source of the dataset. Some dataset may present synthetic data which can significantly affect the outcome of this project. In order to minimize this risk, I made sure to check the site as well as the author of the dataset by assessing their history, the usability rating of their datasets, as well as reconciling commentary of other authors to notebooks that made use of the data.

Conclusion

The process of procuring data is not as simple as it seems. While there are numerous ways to retrieve such repositories in the present due to access through the internet, this open transaction framework gives way for false and misleading data to also enter circulation. In this task, I learned the importance of first verifying your source for this serves as the foundation from which all other preceding steps rely on. Fail in this step and you may compromise every other achievement and render them incorrect.